Fed-HDDH: A Hyperbolic Directional Diffusion Hypernetwork for Federated Learning

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Abstract

The Diffusion Hypernetwork (DH) integrates diffusion-based optimization with hypernetwork-driven parameter generation, offering a powerful paradigm for modeling complex and structured data. However, in heterogeneous distributed settings such as federated learning, existing DH frameworks face fundamental limitations. First, they rely on a single perturbation mechanism, which fails to jointly accommodate heterogeneous modalities — continuous data typically require Gaussian noise, while discrete data depend on auxiliary latent transformations — hindering unified diffusion modeling. Second, we show that hypernetwork-generated weights exhibit severe geometric anisotropy in hyperbolic space under heterogeneous federation; ignoring this anisotropy leads to distorted gradients and unstable optimization, a challenge largely overlooked by prior DH methods. To address these issues, we propose Fed-HDDH, a **Fed**erated **H**yperbolic **D**irectional **D**iffusion **H**ypernetwork. Fed-HDDH treats model weights as diffusion objects in hyperbolic space and introduces a radial–angular constrained forward diffusion that preserves intrinsic anisotropy while enabling cross-client conditional reverse denoising. Extensive experiments across diverse modalities demonstrate that Fed-HDDH consistently outperforms both Euclidean and hyperbolic baselines by an average of 5.7% and improves hypernetwork adaptability by 5.2%.

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CCS Concepts

• **Computing methodologies → Machine learning**.

Keywords

Hypernetwork, Diffusion, Hyperbolic, Federated Learning

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1 Introduction

The Diffusion Hypernetwork (DH) is an emerging generative framework that effectively combines diffusion-based optimization [22] with hypernetwork-driven model generation [18, 64]. By using conditioning signals like task descriptors, DH dynamically generates tailored network weights, allowing fine-grained control over downstream tasks. Diffusion-based hypernetworks outperform conventional ones in expressivity and adaptability, especially for high-dimensional or complex systems [48]. This capability has generated significant interdisciplinary interest. For example, in NLP [33], DH

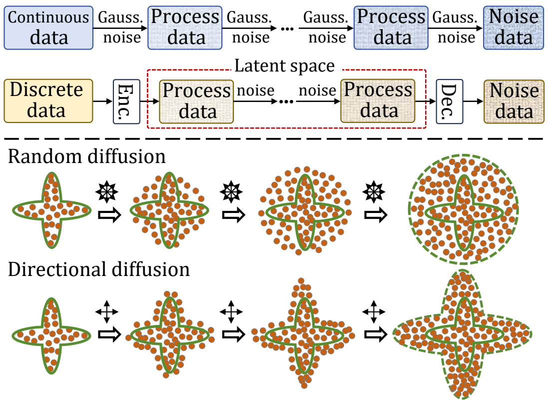


Figure 1: Top: Continuous and discrete diffusion processes. Bottom: Random and directional diffusion.

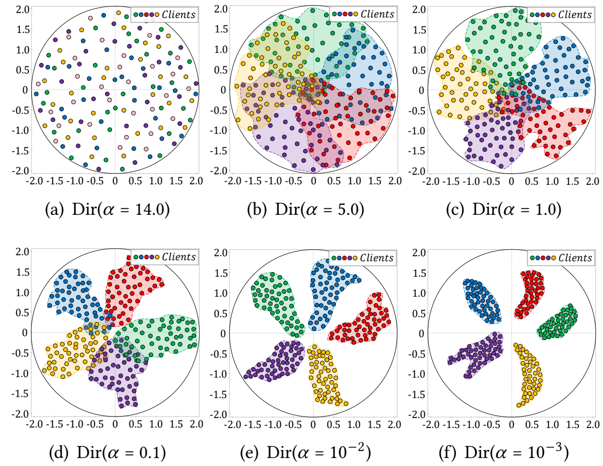


Figure 2: Poincaré disk visualization of target network weights generated by a hypernetwork trained on heterogeneous ImageNet.

significantly improves text generation via enhanced semantics and style control; similarly, in graph neural networks [55], diffusion-based approaches lead to more robust graph representations and superior performance on subsequent tasks like classification or link prediction. Nevertheless, this introduces a major challenge for decentralized implementations such as federated learning, where data heterogeneity and privacy constraints severely hinder the training of diffusion-driven hypernetworks.

i) Limitations in Multimodal Compatibility. Despite its strong generative capacity, DH struggles to accommodate heterogeneous data modalities due to fundamental mismatches in diffusion formulations. As illustrated in Fig. 1(top), continuous data (e.g., image or audio) are typically perturbed by Gaussian noise and reconstructed via reverse denoising [45], whereas discrete modalities (e.g., graph or text) often rely on latent-space mappings or continuous relaxations to enable diffusion [3]. Such discrepancies make it difficult for DH to adopt a unified perturbation mechanism across modalities. The challenge is further amplified in distributed settings such as federated learning [63], where cross-client distribution shifts exacerbate data heterogeneity [62]. Consequently, DH optimization can become unstable, leading to slower convergence and degraded performance across clients [42].

ii) Lack of Directional Optimization. Empirical analysis in hyperbolic space reveals that federated data heterogeneity induces systematic inconsistencies in DH-generated weight distributions. As shown in Fig. 2, training DH on ImageNet [9] under increasing Dirichlet heterogeneity [23] produces projected target network weights in the Poincaré disk, exhibiting clear separability and pronounced directional anisotropy. Such anisotropy encodes intrinsic geometric variations across clients and reflects DH's adaptation to heterogeneous data. Prior studies on hyperbolic optimization reveal that preserving directional structures is critical to mitigate gradient distortion and ensure training stability [46]. However, existing DH frameworks [47] rely on isotropic Gaussian perturbations during diffusion, which rapidly degrade the signal-to-noise ratio and erase these directional cues [71] (see the bottom of Fig. 1). DH thus fails to exploit the anisotropic geometry induced by heterogeneous clients, limiting its adaptability in decentralized learning settings.

To address these challenges, we propose Fed-HDDH, a **Fed**erated **H**yperbolic **D**irectional **D**iffusion **H**ypernetwork. Unlike existing DH frameworks, Fed-HDDH treats target network weights – regardless of data modality – as diffusion entities embedded in hyperbolic space and introduces a radial–angular constrained diffusion mechanism that preserves intrinsic geometric structures. Each client maps locally trained target network (TN) weights into hyperbolic space to obtain geometry-aware diffusion-guiding representations, which undergo a multi-step forward diffusion with jointly constrained radial and angular components. During reverse denoising, clients optimize their local hypernetworks using noised guiding representations and aggregate hypernetwork parameters across clients at each step. This directional diffusion explicitly captures anisotropic structures in weight distributions, improving the adaptability of the generated TNs under federated heterogeneity. Our main contributions are:

- We identify fundamental limitations of existing DH approaches and, for the first time, reveal the critical role of hyperbolic anisotropy in DH optimization.
- We propose a radial–angular constrained diffusion objective that preserves anisotropic geometric structures, enhancing DH performance, and provide theoretical analyses of convergence, complexity, and privacy guarantees.
- Extensive experiments demonstrate consistent performance gains across diverse multimodal datasets.

2 Related Works

Diffusion Hypernetwork. Recent work combines hypernetworks with diffusion models to enable conditional, efficient, and controllable parameter generation. By leveraging hypernetworks to parameterize diffusion processes or diffusion model components, these approaches support on-the-fly adaptation, reduced fine-tuning cost,

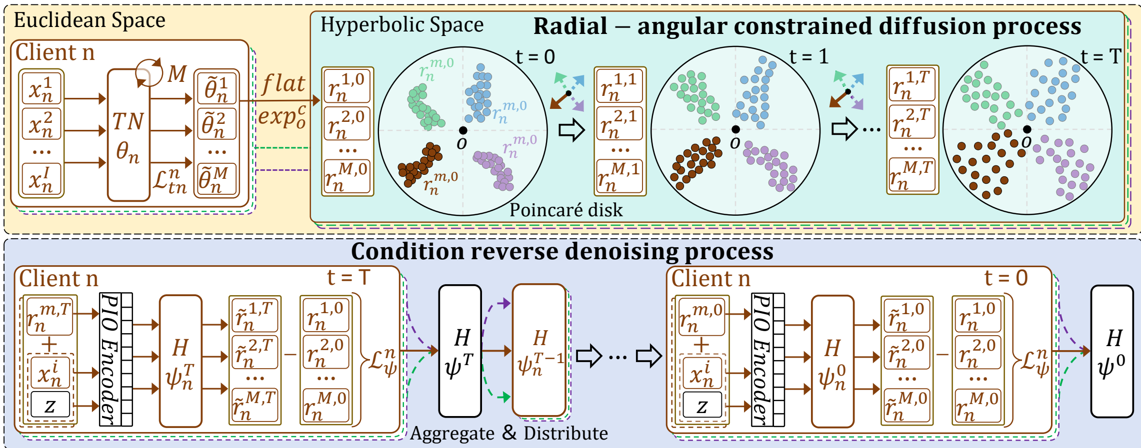


Figure 3: Framework of Fed-HDDH. Clients train TNs and map the resulting weights into hyperbolic guiding representations. Radial–angular constrained forward diffusion injects noise, after which the server applies cross-client reverse denoising to optimize the global hypernetwork.

and improved controllability of generative behavior. A representative line of work focuses on personalization and efficient adaptation. HyperDreamBooth [48] integrates hypernetworks into text-to-image diffusion models to synthesize personalized parameters, substantially reducing training time and storage overhead compared to conventional fine-tuning approaches such as DreamBooth [47] and Textual Inversion [52]. Complementarily, noise hypernetworks amortize diffusion-time optimization by learning noise modulation functions, accelerating sampling while preserving generation quality [13]. Another line of research embeds hypernetworks directly into diffusion architectures to enhance representational flexibility. Hyper-Transforming Latent Diffusion Models employ transformer-based hypernetworks to map latent codes to implicit neural representation (INR) parameters, enabling efficient adaptation without retraining the full diffusion backbone [12, 40]. Similarly, Hyper-DiffusionFields (HyDiF) [4] generate instance-specific neural field weights via a diffusion objective, supporting conditioned generation and molecular inpainting tasks. Hypernetworks also serve as lightweight parameter predictors for diffusion models. HyperLoRA [76] predicts task-specific low-rank adaptation parameters, enabling efficient recontextualization of diffusion models without per-instance fine-tuning. Collectively, these studies demonstrate that diffusion hypernetworks provide a scalable and expressive framework for balancing model capacity, computational efficiency, and controllability across a wide range of generative modeling tasks. However, most existing methods operate in Euclidean parameter spaces with largely isotropic generation processes, lacking geometry-aware or anisotropic modeling of parameter distributions.

3 Methodology

3.1 Problem Formulation

Our federated setup consists of a central server and N clients. Each client n holds a local dataset $\mathcal{D}_n=\{x_n^i\}_{i=1}^I$ drawn from distribution $\mathcal{P}_n$ and trains a randomly initialized target network (TN) $\mathcal{TN}(\cdot;\theta_n)$ for M rounds, producing weights $\{\tilde{\theta}_n^m\}_{m=1}^M$, which are mapped into hyperbolic space to form diffusion-guiding representations $\{r_n^{\mathbb{H},m,0}\}_{m=1}^M$ and undergo a T-step radial–angular constrained forward diffusion to yield noisy states $\{r_n^{\mathbb{H},m,T}\}_{m=1}^M$. During reverse denoising, a client-side hypernetwork $\mathcal{H}(\cdot;\psi_n)$ takes the noisy representation $r_n^{\mathbb{H},m,t}$ and a conditioning signal C=$\{x_n^i, z\}$, where z denotes Gaussian noise, fused via a pretrained Perceiver-IO encoder E_{pio} [26] into a fixed-length embedding h. After each step, the server aggregates and redistributes hypernetwork parameters until convergence. The objective is to learn global ψ such that, given h=$E_{pio}(x_n^i, z)$, the hypernetwork generates TN weights θ_n minimizing the expected loss:

$$\psi^* = \arg\min_{\psi} \frac{1}{NI}\sum_{n=1}^{N}\sum_{i=1}^{I}\mathbb{E}_{x_n^i\sim\mathcal{P}_n}\mathcal{L}_{tn}(x_n^i;\theta_n)\ , \tag{1}$$

where $\mathcal{L}_{tn}$ denotes the TN loss. The hypernetwork output dimension matches the TN parameters count by adjusting the TN architecture or the hypernetwork output layer.

3.2 Diffusion-Guiding Representation

Unlike prior studies that use samples as guidance entities during forward diffusion, Fed-HDDH adopts weight-based guidance derived from the target network (TN). Specifically, each client $n \in [N]$ locally optimizes a randomly initialized TN $\mathcal{TN}(\cdot;\theta_n^E)$ on its dataset $\mathcal{D}_n$=$\{x_n^i\}_{i=1}^I$ for M rounds in the Euclidean space, with the following local training objective:

$$\theta_n^{E,m+1} = \arg\min_{\theta_n^{E,m}} \mathbb{E}_{x_n^i\sim\mathcal{P}_n}\mathcal{L}_{tn}(x_n^i;\theta_n^{E,m}), \tag{2}$$

where $m \in [M]$, $\mathcal{L}_{tn}$ denotes the TN loss and the superscript E indicates Euclidean variables. The TN architecture is modality-dependent (e.g., CNNs for images or GPT-style models for text), while the hypernetwork remains modality-agnostic through this decoupled design [64]. After each round, TN parameters are flattened to obtain representations $r_n^{E,m} = \mathit{flatten}(\theta_n^{E,m})$, forming the set S_n^E =

$\{r_n^{E,m}|m \in [M]\}$, which captures diverse local optima along the optimization trajectory. Each Euclidean representation is then mapped into hyperbolic space via the exponential operator, producing the hyperbolic diffusion-guiding set $S_n^{\mathbb{H}}$:

$$S_n^{\mathbb{H}} = \{r_n^{\mathbb{H},m} = \exp_o^c([0, r_n^{E,m}]) \mid r_n^{E,m} \in S_n^E\}\ , \tag{3}$$

where c denotes the curvature of the hyperbolic space $\mathbb{H}$ and $[0, r_n^{E,m}]$ prepends a zero to ensure the representation lies in the tangent space at the origin. The resulting set $S_n^{\mathbb{H}}$ serves as diffusion-guiding information for client n. While the curvature c affects the embedding geometry, the guidance capability is curvature-invariant, as formalized in Theorem 1.

Theorem 1. ***(Curvature Equivalence of Hyperbolic Representations)***. *Given a representation set $S=\{r^m\}_{m=1}^M$ in a d-dimensional hyperbolic space $\mathbb{H}^{d,c}$ with curvature $-1/c^2 < 0$, there exists, for any $\mathbb{H}^{d,c'}$ with curvature $-1/c'^2 < 0$, a corresponding set $\tilde{S}$ such that $S \xleftrightarrow{equiv.} \tilde{S}$. The equivalence is induced by a curvature-dependent mapping determined solely by c and c'.*

3.3 Hyperbolic Directional Diffusion Process

3.3.1 Geometric Constrained Forward Process. Following standard diffusion optimization, Fed-HDDH performs forward diffusion and reverse denoising on hypernetwork in hyperbolic space, initialized from the guiding representations in Eq. 3. However, isotropic diffusion rapidly degrades the signal-to-noise ratio and distorts the anisotropic structure of the representations, impairing hypernetwork performance [71]. Motivated by recent advances in hyperbolic diffusion [72], we introduce a directional diffusion process that jointly constrains radial and angular components, preserving the anisotropic topology of the generated target network weights.

Radial Constraint. Fed-HDDH adopts the following formulation to preserve curvature-induced radial expansion:

$$\begin{aligned} q\big(r_n^{\mathbb{H},m,t}\big|r_n^{\mathbb{H},m,t-1}\big) &= \sqrt{\overline{\alpha_t}} \cdot r_n^{\mathbb{H},m,t-1} + \sqrt{1-\overline{\alpha_t}} \cdot \epsilon \\ &+ \xi \cdot \tanh\big(1/(\sqrt{c} \cdot \|r_n^{\mathbb{H},m,t-1}\|_2 \cdot k)\big) \cdot r_n^{\mathbb{H},m,t-1}\ , \end{aligned} \tag{4}$$

where $n \in [N]$, $m \in [M]$, and $t \in [T]$ index clients, guidance representations, and diffusion steps, respectively. The injected noise $\epsilon \sim \mathcal{N}(0, I)$. c denotes the hyperbolic curvature, while ξ and k control the diffusion strength and growth rate, adapting to the geometric expansion of hyperbolic space [72]. $\|\cdot\|_2$ denotes the L_2 norm, and $\overline{\alpha_t} := \prod_{i=1}^t (1-\beta_i) \in (0,1)$ defines the variance schedule with a monotonically decreasing sequence $\{\beta_{1:T} \in (0,1)\}$.

Angular Constraint. We define a reference direction from each guidance representation toward the hyperbolic origin to guide the forward diffusion. This strategy has been shown effective in prior hyperbolic studies [14]. Specifically, for the m-th guidance representation of client n, the angular component is defined as:

$$\mathcal{A}_n^m = \mathrm{sign}\big(\log_o^c(r_n^{\mathbb{H},m})\big) \odot (\mu + \sigma \odot \epsilon)\ , \tag{5}$$

where $\log_o^c(\cdot)$ maps the guidance representation to the tangent space at the hyperbolic origin. $\mathrm{sign}(\cdot)$ extracts the sign of the tangent vector, and $\odot$ denotes the Hadamard product. μ and σ denote the mean and standard deviation of the representations. Replacing ϵ in Eq. 4 with this angular component gives the final form of the directional diffusion process:

$$\begin{aligned} q(r_n^{\mathbb{H},m,t}\big|r_n^{\mathbb{H},m,t-1}) &= \sqrt{\overline{\alpha_t}} \cdot r_n^{\mathbb{H},m,t-1} + \sqrt{1-\overline{\alpha_t}} \cdot \mathcal{A}_n^m + \\ &\xi \cdot \tanh\big(1/(\sqrt{c} \cdot \|r_n^{\mathbb{H},m,t-1}\|_2 \cdot k)\big) \cdot r_n^{\mathbb{H},m,t-1}\ . \end{aligned} \tag{6}$$

Each client applies radial–angular constrained diffusion on hyperbolic guidance representations. Theorem 2 shows federated diffusion asymptotically matches its centralized counterpart, justifying distributed hypernetwork optimization.

Theorem 2. ***(Asymptotic Centralization of Federated Diffusion)*** *Let $r_n^{\mathbb{H},m,t}$ denote the m-th guidance representation of client n at diffusion step t, evolving under the geometry-constrained diffusion in Eq. 6. If the dynamics preserve directional alignment with the initialization $r_n^{\mathbb{H},m,0}$, then*

$$\lim_{t\to\infty} r_n^{\mathbb{H},m,t} \xrightarrow{d} |Z_\infty|, \quad Z_\infty \sim \mathcal{N}\Big(\xi\, r_n^{\mathbb{H},m,0}, I\Big),$$

where ξ is determined by the diffusion schedule. Consequently, the empirical distribution over clients converges to a finite mixture of folded normals, and their aggregation converges to a Gaussian. Thus, the federated diffusion is asymptotically equivalent in distribution to centralized diffusion.

3.3.2 Conditional Reverse Denoising Process. Given the noised guidance representations from geometry-constrained forward diffusion (Eq. 6), Fed-HDDH performs conditional reverse denoising to update the hypernetwork:

$$\begin{aligned} &p_{\psi_n}(r_n^{\mathbb{H},m,t-1}\Big|r_n^{\mathbb{H},m,t}, C) = \mathcal{N}(r_n^{\mathbb{H},m,t-1}; \mu_{\psi_n}(r_n^{\mathbb{H},m,t}, t, C), \\ &\Sigma_{\psi_n}(r_n^{\mathbb{H},m,t}, t, C)),\ t \in [T]\ , \end{aligned} \tag{7}$$

where ψ_n denotes the client-specific hypernetwork parameters. The conditioning signal C is a fixed-length embedding constructed by encoding a local data sample x and Gaussian noise z using a pretrained Perceiver-IO encoder $E_{pio}(\cdot)$ [26], i.e., $C = E_{pio}(x, z)$. While the forward diffusion runs locally on each client, reverse denoising is performed in a federated manner. After each denoising step, client-specific hypernetwork parameters ψ_n^t are aggregated via Eq. 8, and the updated global parameters ψ^{t-1} are broadcast to all clients for the next iteration.

$$\{\psi_n^t\}_{n=1}^N \xrightarrow{Aggregate} \psi^{t-1} \xrightarrow{Distribute} \{\psi_n^{t-1}\}_{n=1}^N\ . \tag{8}$$

As a result, the hypernetwork outputs a hyperbolic representation $r_n^{\mathbb{H}}$ conditioned on the mixed sample–noise input of client n, which is mapped back to Euclidean space via $\log_o^c(\cdot)$ to yield the target network weights θ_n^E.

$$\theta_n^E = unflatten\big(\log_o^c(r_n^{\mathbb{H}})\big),\ \forall n \in [N]\ . \tag{9}$$

The optimization procedures are given in Algorithms 1 and 2.

3.4 Optimization

Fed-HDDH adopts an optimization strategy for the hypernetwork similar to standard diffusion models [22], with the key difference that it directly predicts the clean guidance representation $r^{\mathbb{H}}$ at each denoising step, rather than estimating noise. The loss function

Algorithm 1 Training of Fed-HDDH

Input: N: number of clients, $\{\mathcal{D}_n\}_{n=1}^{N}$: training dataset, ψ/θ: Hypernetwork and target network (TN) weights, M: number of guidance representations, T: diffusion steps, η: learning rate.

Output: the hypernetwork $\mathcal{H}(\cdot;\psi^0)$.

Initializing the fixed variance schedule $\{\overline{\alpha}_t\}_{t=1}^{T}$.

```
for n = 1, ..., N do
    Randomly initialize the TN 𝒯𝒩(·; θ_n),
    {θ̃_n^m}_{m=1}^M ← 𝒯𝒩(·; θ_n), optimize TN for M rounds using
    local dataset 𝒟_n (Eq. (2)),
    {r_n^{ℍ,m} = exp_o^c([0, flatten(θ̃_n^m)])}_{m=1}^M, compute initial
    diffusion-guiding representations (Eq. (3))
    q(r_n^{ℍ,m,t} | r_n^{ℍ,m,t−1}) ← Eq. (6), directional diffusion,
end for
for t = T, ..., 1 do
    for n = 1, ..., N do
        Predict the guidance representations {r̃_n^{ℍ,m,t}}_{m=1}^M,
        Compute the loss ℒ_ℋ(ψ_n^t) ← Eq. (10),
        Update ψ_n^t ← ψ_n^t − η∇ψ_n^t,
    end for
    {ψ_n^{t−1}}_{n=1}^N ← Eq. (8), aggregate and distribute the updated
    hypernetwork weights,
end for
return ψ^0.
```

Algorithm 2 Sampling of Fed-HDDH (weight generation)

Input: $\mathcal{D} = \left\{ \bigcup_{n=1}^{N} \mathcal{D}_n \mid \mathcal{D}_n = \{x_n^i \in \mathbb{R}^d\}_{i=1}^{I} \right\}$: training dataset of all clients, T: diffusion steps, $\mathcal{H}(\cdot;\psi)$: optimized hypernetwork.

Output: The generated target network (TN) weights $\tilde{\theta}$.

```
sample x_T ∼ 𝒟, and z ∼ 𝒩(0, I),
h_T = E_pip(x_T, z), mixed encoding using pretrained encoder
E_pio(·),
for t = T, ..., 1 do
    predict r̃^{ℍ,0} with r̃^{ℍ,t}, t and h_t,
    denoise and predict r̃^{ℍ,t−1} ← Eq. (7),
end for
θ̃ = unflatten(log_o^c(r̃^{ℍ,0})) ← Eq. (9), convert the r̃^{ℍ,0},
return θ̂.
```

$\mathcal{L}_{\mathcal{H}}(\psi)$ includes a reconstruction term $\|\mathcal{H}(h;\psi) - r^{\mathbb{H}}\|_2$ measuring the discrepancy between the hypernetwork output and the target guidance representation. To account for random noise in the conditioning input, we further encourage an idempotence-like behavior [64], meaning repeated applications of $\mathcal{H}(\cdot;\psi)$ produce consistent outputs. This allows the hypernetwork to refine its prediction adaptively based on its previous output. Specifically, we introduce an auxiliary consistency term $\|\mathcal{H}(\mathcal{H}(h;\psi);\hat{\psi}) - \mathcal{H}(h;\psi)\|_2$, where $\hat{\psi}$ is a frozen copy of ψ (gradients do not update $\hat{\psi}$), a strategy shown effective in prior work [51]. The full loss is thus:

$$\begin{aligned}\mathcal{L}_{\mathcal{H}}(\psi) =&(1-\lambda)\left\|\mathcal{H}(h;\psi) - r^{\mathbb{H}}\right\|_2 + \lambda\left\|\mathcal{H}(\mathcal{H}(h;\psi);\hat{\psi}) - \right. \\ &\left. \mathcal{H}(h;\psi)\right\|_2 ,\end{aligned} \tag{10}$$

where $\lambda \in (0,1)$ balances the two loss terms.

3.5 Convergence and Complexity Analysis

3.5.1 Convergence Analysis. Consider the reverse denoising process (Eq. 7), where $\mathcal{H}(\cdot;\psi_n^t)$ is the local hypernetwork on client n at step t, and $\{r_n^t\}_{t=1}^{T}$ the sequence of denoised guidance representations. Assume $\mathcal{H}(\cdot;\psi_n^t)$ is L_n^t-Lipschitz with $L_n^t < 1$, which can be ensured via ReLU activations, spectral normalization, or a global output scaling [53]. The noise ϵ^t is zero-mean with bounded magnitude $\mathbb{E}[\|\epsilon^t\|] \le \delta_t$, where δ_t may decay along the denoising schedule. After aggregation, the global hypernetwork $\mathcal{H}(\cdot;\psi^t) = \frac{1}{N}\sum_{n=1}^{N}\mathcal{H}(\cdot;\psi_n^t)$ is L^t-Lipschitz with $L^t = \frac{1}{N}\sum_{n=1}^{N} L_n^t < 1$ [36]. By the Banach fixed-point theorem [25], it admits a unique fixed point r_t^* satisfying $\mathcal{H}(r_t^*;\psi^t) = r_t^*$. With update rule $r^{t+1} = \mathcal{H}(r^t;\psi^t) + \epsilon^t$, the distance to the fixed point satisfies $\mathbb{E}[\|r^{t+1} - r_t^*\|] \le L^t\mathbb{E}[\|r^t - r_t^*\|] + \delta_t$, and iterating yields

$$\mathbb{E}\left[\|r^t - r_t^*\|\right] \le \left(\prod_{k=0}^{t-1} L^k\right)\|r^0 - r_0^*\| + \sum_{j=0}^{t-1}\left(\prod_{k=j+1}^{t-1} L^k\right)\delta_j .$$

Thus, if $\sup_t L^t \le L < 1$ and $\delta_t \to 0$, then $\lim_{t\to\infty}\mathbb{E}[\|r^t - r_t^*\|] = 0$. In summary, under Lipschitz continuity, bounded and decaying noise, and client aggregation, the hypernetwork in Fed-HDDH converges in expectation to the unique fixed point.

3.5.2 Complexity Analysis. Each client $n \in [N]$ holds I training samples of dimension d'. Auxiliary operations such as mixed encoding and parameter aggregation are omitted for clarity. **Time Complexity:** Training includes three components: (1) Guidance representation optimization (Eq. 2): $O(d'IMNP_{tn})$, where M is the number of representations per client and P_{tn} is the cost of a single target network forward pass. (2) Directional diffusion (Eq. 6): $O(dMNT)$, with d the representation dimension and T the number of diffusion steps. (3) Reverse denoising (Eq. 7): $O(dMNTP_h)$, where P_h is the cost of a hypernetwork forward pass. Overall, the total training-time complexity is $O\big(MN(d'IP_{tn} + dT + dTP_h)\big)$. **Space Complexity:** Each client maps M guidance representations $r^{E,m}, m \in [M]$ into d-dimensional hyperbolic vectors, yielding MN hyperbolic representations $r^{\mathbb{H},m}$. Thus, the total space complexity is $O(dMN)$.

3.6 Privacy Guarantee of Fed-HDDH

Unlike standard federated learning, Fed-HDDH never transmits gradients from local target network optimization (Eq. 2), fundamentally reducing the risk of training-sample leakage. During reverse denoising (Eq. 7), only hypernetwork parameter updates are exchanged. As a result, even under strong adversarial attacks — including model inversion [24], membership inference [39], and gradient-leakage attacks [75] — the hypernetwork parameters reveal little about individual clients' private data. Inspired by central and local differential privacy [58], we introduce **d**iffusion **d**ifferential **p**rivacy (DDP), a privacy notion tailored to diffusion-guided hypernetworks, and provide theoretical analysis showing that Fed-HDDH satisfies rigorous privacy guarantees.

Definition 1. ***((v,ρ)-CDDP).*** *A hypernetwork $\mathcal{H}(\cdot;\psi)$ satisfies (v,ρ)-center diffusion differential privacy if, for any two adjacent input sets $\mathcal{R} = \{r_1,\ldots,r_m\}$ and $\mathcal{R}' = \{r_1',\ldots,r_m'\}$ differing in at most one element ($|\mathcal{R}\triangle\mathcal{R}'| = 1$), and for any measurable output set*

Table 1: Comparison of classification accuracy (%) on continous image datasets. (Bold: best; <u>Underline</u>: runner-up.)

Grp.	Methods	TinyImageNet			CIFAR-100			CIFAR-10		
		Dir(α=10)	Dir(α=0.5)	Dir(α=0.01)	Dir(α=10)	Dir(α=0.1)	Dir(α=0.01)	Dir(α=10)	Dir(α=0.1)	Dir(α=0.01)
Euclidean	MOON	49.75 ± 0.71	48.03 ± 0.93	47.23 ± 0.55	67.53 ± 0.27	65.43 ± 1.13	63.74 ± 0.43	55.76 ± 0.36	54.14 ± 0.12	53.53 ± 0.41
	FedDyn	47.62 ± 0.53	46.13 ± 0.43	45.01 ± 0.11	55.57 ± 0.13	54.13 ± 0.63	53.32 ± 0.21	83.31 ± 0.51	82.03 ± 0.22	80.31 ± 0.44
	FedPerfix	50.32 ± 0.13	49.13 ± 0.11	48.51 ± 0.32	55.41 ± 0.82	54.83 ± 0.25	53.04 ± 0.66	67.45 ± 0.51	66.53 ± 0.32	65.82 ± 0.45
	FedMKD	51.76 ± 0.43	50.86 ± 0.81	50.01 ± 1.01	56.03 ± 0.25	55.43 ± 1.02	54.01 ± 0.31	65.36 ± 0.62	63.78 ± 0.02	61.62 ± 0.43
	i-MAE	61.59 ± 0.73	60.23 ± 0.21	58.44 ± 0.72	69.67 ± 0.25	67.97 ± 1.02	65.34 ± 0.51	79.94 ± 0.41	77.14 ± 0.83	76.13 ± 1.21
	SANE	64.21 ± 0.42	62.22 ± 0.81	60.04 ± 0.44	72.87 ± 0.25	70.81 ± 0.34	68.23 ± 0.51	88.42 ± 0.12	86.23 ± 1.32	84.22 ± 0.61
Hyperbolic	ClippedHC	64.87 ± 0.07	62.13 ± 0.13	60.86 ± 0.16	77.14 ± 0.05	75.11 ± 0.04	73.71 ± 0.35	88.54 ± 0.32	85.21 ± 0.43	82.13 ± 0.62
	PoincaréRN	62.01 ± 0.48	60.87 ± 0.71	58.31 ± 0.04	76.63 ± 0.29	74.25 ± 1.14	72.45 ± 0.19	85.21 ± 0.11	83.21 ± 0.55	80.41 ± 0.63
	FedMRUR	**65.71** ± 0.33	<u>63.78</u> ± 1.32	<u>61.73</u> ± 1.07	65.93 ± 0.23	64.23 ± 0.73	63.43 ± 0.31	77.32 ± 0.11	76.14 ± 0.52	74.75 ± 1.31
	FedHyper	62.42 ± 0.52	60.95 ± 0.81	59.21 ± 0.31	60.95 ± 0.12	59.24 ± 0.11	57.35 ± 1.44	75.82 ± 0.19	73.82 ± 0.13	70.42 ± 0.53
	HCNN	<u>65.64</u> ± 0.43	63.11 ± 0.53	59.13 ± 0.82	**78.13** ± 0.21	<u>76.43</u> ± 0.62	<u>74.23</u> ± 0.45	**89.55** ± 0.51	<u>87.42</u> ± 0.42	<u>84.24</u> ± 0.81
	LViT	64.24 ± 0.04	61.43 ± 0.74	60.02 ± 0.05	69.85 ± 0.03	68.63 ± 0.15	67.24 ± 0.32	85.14 ± 0.52	83.53 ± 0.89	80.04 ± 0.95
	Fed-HDDH	65.51 ± 0.31	**65.34** ± 0.41	**65.43** ± 1.22	<u>78.03</u> ± 0.15	**77.84** ± 0.44	**77.93** ± 0.33	<u>89.22</u> ± 0.11	**88.94** ± 0.31	**89.05** ± 0.41

O, it holds that

$$\mathbb{P}\big[\mathcal{H}(\mathcal{R};\psi) \in O\big] \le e^{\upsilon}\, \mathbb{P}\big[\mathcal{H}(\mathcal{R}';\psi) \in O\big] + \rho.$$

Definition 2. ***((υ, ρ)-LDDP).*** *A hypernetwork $\mathcal{H}(\cdot;\psi)$ satisfies (υ, ρ)-local diffusion differential privacy if, for any inputs r, r' and any measurable output set O,*

$$\mathbb{P}\big[\mathcal{H}(r;\psi) \in O\big] \le e^{\upsilon}\, \mathbb{P}\big[\mathcal{H}(r';\psi) \in O\big] + \rho.$$

where υ is the privacy budget (smaller values imply stronger privacy) and ρ bounds privacy leakage probability.

The core idea of (υ, ρ)-CDDP and (υ, ρ)-LDDP is to ensure that hypernetwork outputs are statistically indistinguishable with respect to client- or sample-level participation in the diffusion-based optimization. This indistinguishability bounds an adversary's ability to infer participation from the hypernetwork outputs. Compared to CDDP, (υ, ρ)-LDDP provides a strictly stronger guarantee by protecting each individual sample rather than an entire client. In Theorem 3, we show that Fed-HDDH satisfies the LDDP property.

Theorem 3. ***(Privacy Guarantee of Fed-HDDH).*** *Given the guidance representation set $\{r_n^m\}_{n\in[N]}^{m\in[M]}$, if we choose the conditional denoising step to be T, noise scheduling parameters to be $\{\beta_t\}_{t=1}^{T}$, then for any $\rho \in (0, 1)$, the output of the hypernetwork $\mathcal{H}(\cdot;\psi)$ satisfies (υ, ρ)-LDDP with*

$$\upsilon \le C_{geo}\left(\frac{2\alpha_0 D_{max}^2}{1-\alpha_{tot}} + D_{max}\sqrt{\frac{8\,\alpha_0\, \log(1/\rho)}{1-\alpha_0}}\right),$$

where D_{max} bounds the geodesic distance between adjacent client representation embeddins, α_0 is the forward diffusion scale, α_{tot} is the total residual diffusion factor, and C_{geo} is a geometry-dependent constant.

4 Experiments

Datasets. We evaluate Fed-HDDH on both continuous and discrete modalities. Continuous datasets include images (TinyImageNet [28], CIFAR-100/10 [27]), videos (XD-Violence [67], UCF-Crime [54]), and domain adaptation benchmarks (PACS [31], DomainNet [41], Office-Home [21]). Classification accuracy is used unless otherwise noted (frame-level *AP* for XD-Violence, *AUC* [29] for UCF-Crime). Discrete modalities include graph datasets PubMed, CiteSeer, and Airport [6], with node classification accuracy. Data are partitioned across clients via Dirichlet(α) sampling to control heterogeneity, where a smaller α indicates stronger non-IID settings.

Baselines. We compare Fed-HDDH with representative *Euclidean* and *Hyperbolic* baselines across multiple modalities. **Image:** Euclidean/Hyperbolic baselines include MOON [32], FedDyn [11], FedPerfix [56], FedMKD [57], i-MAE [73], SANE [50]/ClippedHC [17], PoincaréRN [59], FedMRUR [2], FedHyper [34], HCNN [5], LViT [19]. **Video:** Euclidean/Hyperbolic baselines include FedPPVU [10], FedCLAP [1], FedCoOp [16], FedVAD [43], WSVAD [60]/WSRGB [44], DSRL [29], PierEye [30]. **Graph:** Euclidean/Hyperbolic baselines include SGC [66], Fed-Sage [74], GraphCON [49], NodeFormer [68], ACMP [65], GraphFL [61], SGFormer [69], FedSCem [70]/HGCL [35], F-HNN [7], H-GRAM [8], LRN [20], H-EDML [6]. Non-federated baselines are adapted with FedAvg algorithm [37].

Implementation Details. We adopt an MLP as the hypernetwork backbone with ReLU activations [15], apply spectral normalization [38] to all layers, and introduce a global output scaling factor $s \in (0, 1)$. Across modalities, the target network (TN) architecture and loss $\mathcal{L}_{tn}$ follow the corresponding baselines, with the hypernetwork output dimension matching the TN parameters. By default, we set the client number N, hypernetwork input dimension d, guidance representation number M, diffusion step number T, diffusion step size ξ, diffusion growth rate k, noise length l_z, learning rate η, loss coefficient λ, and hyperbolic curvature to 10^2, 10^5, 10^2, 10^3, 0.3, 0.01, 10^3, 10^{-3}, 0.5, and -1, respectively. The noise schedule $\beta_{1:T}$ is bounded in $[10^{-3}, 10^{-1}]$. All experiments are implemented in *PyTorch* and trained on a single *NVIDIA Tesla V100*.

4.1 Evaluation of Continuous and Discrete Modalities

Image Classification. We evaluate image classification on TinyImageNet and CIFAR-100/10 under various heterogeneity levels. Table 1 reports mean accuracy and standard deviation over three runs. Overall, Fed-HDDH outperforms Euclidean and hyperbolic baselines by 16.6% and 4.4% on average. Under near i.i.d. settings (α=10), Fed-HDDH remains competitive, though not always the top performer. As heterogeneity increases (smaller α), non-i.i.d.-oriented baselines such as MOON, FedPerfix, and FedHyper degrade noticeably, while others deteriorate more severely. In contrast, Fed-HDDH

Table 2: Comparison of node classification accuracy (%) on discrete graph datasets. (Bold: best; Underline: runner-up.)

Grp.	Methods	PubMed (δ-hyperbolicity = 3.5)			CiteSeer (δ-hyperbolicity = 5)			Airport (δ-hyperbolicity = 1)		
		Dir(α=5)	Dir(α=0.1)	Dir(α=0.01)	Dir(α=5)	Dir(α=0.1)	Dir(α=0.01)	Dir(α=2)	Dir(α=0.1)	Dir(α=0.01)
Euclidean	SGC	78.31 ± 0.04	76.23 ± 0.12	74.21 ± 0.41	72.01 ± 0.51	70.31 ± 0.47	68.63 ± 0.21	83.85 ± 1.12	81.43 ± 0.16	79.51 ± 0.47
	FedSage	80.04 ± 0.14	78.37 ± 0.82	76.83 ± 0.13	71.35 ± 0.63	69.78 ± 0.13	67.83 ± 0.47	81.78 ± 0.73	79.74 ± 0.31	77.03 ± 1.02
	GraphCON	78.79 ± 0.81	76.67 ± 0.51	74.78 ± 1.13	71.12 ± 0.34	69.21 ± 0.81	67.83 ± 0.53	80.11 ± 0.21	78.57 ± 1.41	76.54 ± 0.61
	NodeFormer	79.07 ± 0.71	77.31 ± 1.45	76.01 ± 0.23	72.61 ± 0.11	70.42 ± 0.51	68.64 ± 0.64	82.83 ± 1.12	80.45 ± 0.24	78.24 ± 1.03
	ACMP	78.47 ± 0.04	76.32 ± 0.76	74.32 ± 0.61	73.11 ± 0.44	71.53 ± 0.12	69.49 ± 0.42	88.92 ± 1.12	86.13 ± 0.05	84.91 ± 1.87
	GraphFL	79.96 ± 0.34	77.32 ± 0.83	75.42 ± 0.81	67.42 ± 0.42	66.42 ± 0.12	65.97 ± 0.42	80.84 ± 0.71	78.49 ± 0.15	77.74 ± 0.51
	SGFormer	79.31 ± 0.26	77.23 ± 0.54	75.53 ± 0.52	72.25 ± 0.46	70.51 ± 0.47	68.63 ± 0.64	91.63 ± 1.02	89.42 ± 0.75	87.67 ± 1.25
	FedSCem	**85.06** ± 0.04	83.51 ± 0.13	81.23 ± 0.44	**84.03** ± 0.44	82.71 ± 0.73	80.49 ± 0.61	89.68 ± 1.12	87.73 ± 0.02	84.73 ± 1.07
Hyperbolic	HGCL	79.34 ± 0.35	78.83 ± 0.71	77.52 ± 0.67	72.24 ± 0.12	71.03 ± 0.24	70.15 ± 0.51	92.31 ± 1.01	91.59 ± 0.77	90.17 ± 0.97
	F-HNN	77.62 ± 0.89	76.84 ± 1.05	75.56 ± 0.78	71.08 ± 0.18	70.51 ± 0.65	69.65 ± 0.16	91.83 ± 0.71	90.52 ± 0.05	89.62 ± 0.81
	H-GRAM	78.01 ± 0.68	77.53 ± 0.51	76.91 ± 0.41	72.12 ± 0.51	71.21 ± 0.61	70.18 ± 0.12	89.81 ± 1.12	88.15 ± 0.65	87.61 ± 0.81
	LRN	77.20 ± 0.31	76.18 ± 0.06	75.71 ± 0.74	66.97 ± 1.78	65.56 ± 0.17	64.35 ± 0.43	92.36 ± 1.22	91.58 ± 1.61	90.71 ± 0.37
	H-EDML	79.28 ± 0.21	78.89 ± 0.41	77.43 ± 0.19	74.12 ± 1.02	73.78 ± 0.52	72.31 ± 0.53	93.71 ± 0.42	92.74 ± 0.46	91.52 ± 0.54
	Fed-HDDH	84.93 ± 0.31	**84.82** ± 0.38	**84.91** ± 0.53	83.85 ± 0.82	**83.92** ± 1.07	**83.73** ± 0.14	**95.63** ± 1.14	**95.78** ± 0.46	**94.89** ± 0.57

maintains consistently strong performance across all heterogeneity levels, demonstrating the robustness of directional diffusion optimization to image-level heterogeneity.

Video Abnormal Detection. We evaluate anomaly detection on XD-Violence and UCF-Crime. Table 3 reports average frame-level AP and AUC over three runs. On XD-Violence, Fed-HDDH outperforms Euclidean and hyperbolic baselines by 9.8% and 2.1%, respectively; on UCF-Crime, the gains are 2.9% and 1.4%. Under near i.i.d. settings (α=5), Fed-HDDH remains competitive, and under severe heterogeneity (α=10^{-2}), it consistently outperforms all baselines. These results demonstrate the robustness of directional diffusion optimization for video anomaly detection under heterogeneous data distributions.

Table 3: Comparison of frame-level AP (%) on XD-Violence and AUC (%) on UCF-Crime datasets.

Grp.	Methods	XD-Violence (AP ↑)		UCF-Crime (AUC ↑)	
		Dir(α=5)	Dir(α=0.01)	Dir(α=5)	Dir(α=0.01)
Euclidean	RTFM	77.62 ± 0.61	73.89 ± 1.43	84.27 ± 0.31	80.32 ± 0.21
	MSL	78.17 ± 0.34	74.34 ± 0.51	85.32 ± 0.13	81.52 ± 0.61
	UMIL	81.67 ± 0.35	77.34 ± 0.35	86.57 ± 0.17	82.23 ± 0.87
	CU-Net	81.38 ± 0.43	77.95 ± 0.42	86.01 ± 1.12	82.62 ± 0.03
	LAVAD	80.04 ± 0.13	75.24 ± 0.52	80.13 ± 0.31	74.56 ± 1.05
	VERA	79.45 ± 0.83	75.21 ± 0.42	86.34 ± 0.27	82.81 ± 0.82
	FedPPVU	85.65 ± 0.34	81.24 ± 0.91	82.94 ± 0.32	78.27 ± 0.84
	FedCLAP	85.45 ± 0.64	81.62 ± 0.31	83.19 ± 0.41	79.85 ± 0.13
	FedCoOp	71.86 ± 0.52	77.94 ± 0.98	84.07 ± 1.07	80.21 ± 0.82
	FedVAD	66.75 ± 0.81	64.92 ± 0.81	85.11 ± 0.61	81.93 ± 0.73
	WSVAD	75.41 ± 0.67	71.73 ± 0.45	84.11 ± 0.26	80.01 ± 0.91
Hyperbolic	WSRGB	82.62 ± 0.32	80.13 ± 0.73	85.21 ± 0.42	83.31 ± 0.33
	DSRL	87.62 ± 0.29	85.52 ± 0.42	86.21 ± 0.32	84.04 ± 0.43
	PierEye	**87.72** ± 0.82	84.27 ± 0.52	**86.63** ± 0.11	83.92 ± 0.43
	Fed-HDDH	87.64 ± 0.21	**87.51** ± 0.63	86.58 ± 0.14	**86.29** ± 1.12

Graph Node Classification. We evaluate Fed-HDDH on PubMed, CiteSeer, and Airport using node classification accuracy, with client heterogeneity controlled by Dirichlet(α) sampling [23]. Table 2 reports results over three runs. Overall, Fed-HDDH outperforms Euclidean and hyperbolic baselines by 9.2% and 2.1% on average. As α decreases, Euclidean and hyperbolic baselines drop by 4.3% and 1.9%, respectively, while Fed-HDDH remains stable, indicating improved robustness to graph heterogeneity.

Adaptability. We evaluate cross-domain adaptability on PACS, DomainNet, and Office-Home. Table 4 reports average cross-domain accuracy. With noise length $l_z = 10^3$, Fed-HDDH achieves state-of-the-art performance on all datasets, outperforming Euclidean and hyperbolic baselines by 5.6% and 4.5% on average. This gain stems from performing diffusion in model parameter space rather than data space, yielding greater robustness to domain shifts. Although extreme l_z values reduce this benefit, Fed-HDDH consistently improves adaptability across all settings.

Table 4: Cross-domain accuracy (%) on adaptation datasets.

Grp.	Methods	PACS	DomainNet	Office-Home
Euclidean	MOON	79.15 ± 0.23	58.18 ± 0.24	80.53 ± 0.85
	FedDyn	80.02 ± 0.12	58.89 ± 0.63	81.65 ± 0.43
	FedPerfix	81.22 ± 0.83	59.72 ± 0.35	82.12 ± 0.64
	FedMKD	81.45 ± 0.42	60.86 ± 0.75	82.42 ± 0.15
	i-MAE	82.73 ± 0.58	60.01 ± 0.32	83.78 ± 0.72
	SANE	82.15 ± 0.81	61.32 ± 0.18	83.84 ± 0.23
Hyperbolic	ClippedHC	81.35 ± 0.85	59.78 ± 0.23	80.54 ± 0.53
	PoincaréRN	81.73 ± 0.74	61.62 ± 0.53	81.96 ± 0.44
	FedMRUR	82.71 ± 0.85	59.36 ± 0.12	82.12 ± 0.76
	FedHyper	83.12 ± 0.25	59.62 ± 0.85	85.24 ± 0.23
	HCNN	83.15 ± 0.26	60.46 ± 0.34	84.19 ± 0.64
	LViT	83.01 ± 0.11	60.87 ± 0.35	83.39 ± 0.24
	Fed-HDDH (l_z=10^2)	84.42 ± 0.31	61.82 ± 0.46	85.78 ± 0.07
	Fed-HDDH (l_z=10^3)	**85.23** ± 0.54	**63.73** ± 0.56	**86.62** ± 0.11
	Fed-HDDH (l_z=10^4)	83.63 ± 0.32	60.53 ± 0.73	84.13 ± 0.51

4.2 Analysis of Fed-HDDH

Comparison of Training Cost. We compare computational cost (GFLOPs, G=10^9) and peak GPU memory usage. Fig. 4 shows the cost distribution across modalities, where solid/dashed circles denote Euclidean/hyperbolic baselines and the star indicates Fed-HDDH. Hyperbolic baselines incur 11.5% higher computation and 15.4% higher memory than Euclidean ones on average, mainly due to nonlinear cross-space operators (e.g., exp). In contrast, Fed-HDDH

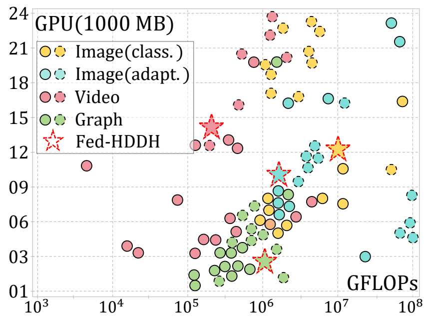
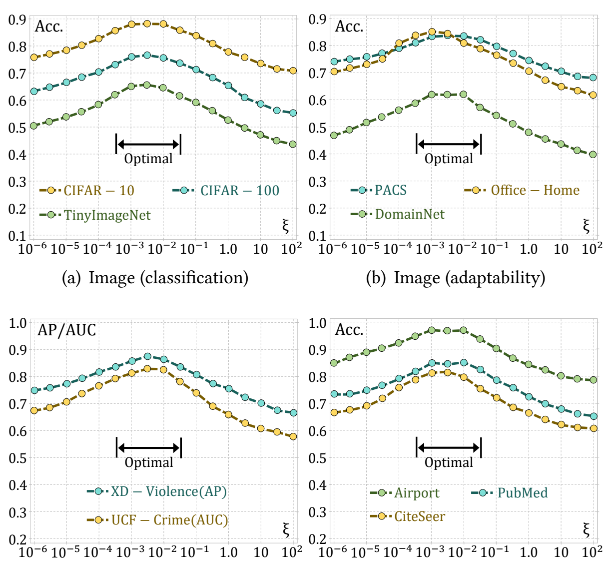


Figure 4: Training costs across different datasets.

increases computation and memory by only 6.3% and 7.8%, ranking among the most efficient hyperbolic methods, as it applies hyperbolic operations only to model parameters rather than samples.

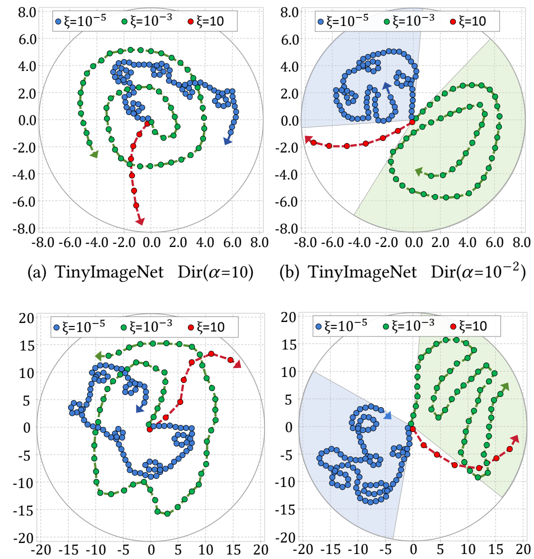
Impact of Diffusion Step Size ξ. Fig. 5 analyzes the effect of the diffusion step size ξ (Eq. 4) on target network performance across modalities. As ξ increases from 10^{-6} to 10^{2}, performance first improves and then degrades, with optimal results in the range $[5\times10^{-4}, 5\times10^{-2}]$. Extremely large ξ (e.g., 10) causes unstable updates, while very small values (e.g., 10^{-6}) slow convergence. Poincaré disk visualizations (Fig. 6) show that i.i.d. settings (larger α) yield broader exploration, whereas heterogeneous settings (smaller α) produce more localized trajectories. Under-diffusion leads to densely packed loops, over-diffusion causes abrupt boundary jumps, and the optimal diffusion ξ=10^{-3} results in smooth, well-spread, directional trajectories, indicating stable and effective learning.



(a) Image (classification) (b) Image (adaptability)

(c) Video (d) Graph

Figure 5: Effect of diffusion step size ξ on different datasets.



(a) TinyImageNet Dir(α=10) (b) TinyImageNet Dir(α=10^{-2})

(c) PubMed Dir(α=5) (d) PubMed Dir(α=10^{-2})

Figure 6: Optimization trajectory of hypernetwork weights.

Ablation Study. We evaluate the effects of the radial (Eq. 4) and angular (Eq. 5) constraint in Fed-HDDH under Dirichlet heterogeneity across modalities. As shown in Table 5, removing the radial (w/o R), angular (w/o A), or both constraints (w/o RA) consistently degrades performance. On PACS, the angular constraint is more critical for adaptability, whereas on other datasets the radial constraint contributes more to overall performance. Overall, the radial constraint primarily improves performance, the angular constraint enhances adaptability, and both are complementary and necessary.

Table 5: Ablation study. (Bold: best; Underline: runner-up.)

R	A	TinyImageNet (*Image*)			XD-Violence (*Video*)		
		Dir(α=10)	Dir(α=0.5)	Dir(α=0.01)	Dir(α=5)	Dir(α=0.1)	Dir(α=0.01)
✓	✓	**65.50** ± 0.1	**65.46** ± 0.3	**65.39** ± 0.1	**87.63** ± 0.4	**87.25** ± 0.3	**87.49** ± 0.2
✓	✗	61.43 ± 0.1	59.89 ± 0.4	59.04 ± 0.3	81.15 ± 0.5	79.43 ± 0.1	72.43 ± 0.2
✗	✓	57.34 ± 0.5	55.43 ± 0.7	53.25 ± 0.2	79.23 ± 1.4	73.13 ± 0.4	66.32 ± 0.2
✗	✗	55.75 ± 0.2	54.56 ± 0.2	51.71 ± 0.5	72.45 ± 0.1	68.35 ± 0.1	64.21 ± 0.3
R	A	PubMed (*Graph*)			PACS (*Adaptability*)		
		Dir(α=5)	Dir(α=0.1)	Dir(α=0.01)	Dir(α=10)	Dir(α=0.1)	Dir(α=0.01)
✓	✓	**84.95** ± 0.1	**84.86** ± 0.1	**84.91** ± 0.3	**85.15** ± 0.3	**85.08** ± 0.2	**85.21** ± 0.1
✓	✗	80.47 ± 0.7	78.54 ± 0.1	73.21 ± 0.4	78.35 ± 1.5	76.25 ± 0.5	68.32 ± 0.3
✗	✓	75.43 ± 0.1	72.54 ± 0.2	62.34 ± 0.1	81.64 ± 0.7	80.09 ± 0.3	79.53 ± 0.1
✗	✗	69.42 ± 1.1	66.23 ± 0.6	59.31 ± 0.3	74.14 ± 0.6	68.62 ± 0.2	60.23 ± 0.6

5 Conclusion

This paper identifies limitations of DH methods in distributed settings and reveals anisotropic behavior in hyperbolic space. We propose Fed-HDDH, a hypernetwork-based hyperbolic directional diffusion framework with theoretical convergence and privacy guarantees. Extensive experiments demonstrate significant gains in performance and adaptability.

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